



Answer the following questions

Question 1:

[20 Marks]

- A) State whether each of the following statements is either True or False. [10 Marks]
1. A non-linearly separable training set in a given feature space can be made linearly separable in another space.
 2. When filtering an image with a Gaussian filter, the kernel size is the most important because it defines the scale.
 3. Adaboost classifier accounts for outliers by lowering the weights of training points that are repeatedly misclassified.
 4. Averaging is considered an intuitive solution for image smoothing as it assumes the independence of the noise added to the image.
 5. Gaussian filter is edge preserving.
 6. When applying Sobel filtering to an image, both correlation and convolution will produce the same results.
 7. The k-NN classifier does not require an explicit training step.
 8. If we initialize the Mean-shift clustering algorithm with the same window size but different starting positions for the window, the algorithm will always converge to the same solution.
 9. When removing an image noise using Gaussian filter, the magnitude of noise decreases with the decrease of the blur kernel's scale σ .
 10. For an SVM, if we remove one of the support vectors from the training set, the size of the maximum margin increases.

B) Answer briefly the following questions:

- i. What is template matching? How it can be applied using filters? [2 Marks]
- ii. What arithmetic operation you can do to sharpen an image? Give an example by using filters. [2 Marks]
- iii. Name two methods for deciding whether two feature descriptors match or not. [2 Marks]
- iv. How many point correspondences are needed to fit a 2D translational model? [1 Mark]
- v. Given the following image as a result of k-means clustering, if you want pixels of similar color but in discontinuous regions to be clustered in different clusters, what kind of feature(s) you may use as input to the k-means algorithm? [1 Mark]



Original image



K-means result with k=3

- vi. **RANSAC** (RANdom Sample Consensus) is an algorithm for fitting a model to very noisy data. Its basic idea is to fit models to small samples of data, and then look for "consensus" from the entire dataset. Complete the following pseudo-code for RANSAC [2 Marks]

INPUT:

data - a set of observations
 model - a model that can be fitted to data
 s - the minimum number of data required to fit the model (fixed for a given model)
 t - a threshold value for determining when a data point fits a model
 d - the number of "consensus" data points required for a "good" model

OUTPUT: BestModel

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FOR (a number of iterations)
  CurrentSet := sample a data subset of size s
  CurrentModel := model fitted to CurrentSet
  ConsensusSet := CurrentSet
  FOR every data point not in CurrentSet
    .....
  END FOR
  IF #(ConsensusSet) > d
    .....
  END IF
END FOR
RETURN BestModel

```

[10 Marks]**Question 2:**

- A) Describe the **Harris** detector for corner detection of a 2D image by giving the main steps of the algorithm with description of the used parameters. **[3 Marks]**
- B) The **SIFT** descriptor is a popular method for describing selected feature points based on local neighborhood properties so that they can be matched reliably across images. Assuming feature points have been previously detected using the **SIFT** feature detector, **[3 Marks]**
- Briefly describe the main step(s) of creating the **SIFT** feature descriptor at a given feature point.
 - Name three image changes that the **SIFT** descriptor is invariant to.
- C) In foreground-background image segmentation, the **Grabcut** is considered one of the fundamental approaches. Answer the following questions about Grabcut: **[4 Marks]**
- What image features can be used for segmentation?
 - How initial segmentation of image pixels can be done?
 - What kind of model representation can be formulated to define both foreground and background classes?
 - What is the main technique upon which the Grabcut approach depends for graph partitioning.

[15 Marks]**Question 3:**

- A) Consider the following two-class data set **[5 Marks]**
- Visualize the given data set on a 2-D plot.
 - If this data is to be classified using a linear **SVM**, which data points do you think should be used as support vectors? Highlight them on your plot.
 - Calculate the equation of the decision boundary learnt by this SVM, and the Margin (**M**) that maximizes the distance between the two classes.

x_1	x_2	t
1	1	-1
0	0	-1
2	1	-1
2	3	1
1	2	1

- B) The following very useful operator $\nabla^2 h_\sigma(x; y)$ is often applied to an image $I(x; y)$ in computer vision algorithms, to generate a related image $g(x; y)$, where $\nabla^2 = (\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2})$ **[6 Marks]**

- i. Give the general name for this mathematical operator, and the chief purpose that it serves in computer vision.
- ii. What image properties should correspond to the zeroes of the equation, i.e. those points $(x; y)$ in the image $I(x; y)$ where the above result $g(x; y) = 0$?
- iii. What is the significance of the parameter σ ? If you increased its value, would there be more or fewer points $(x; y)$ at which $g(x; y) = 0$? And why?
- iv. Explain which of the following two operations is preferable to apply and why? where * signifies 2D convolution operation,
 - a. $\nabla^2 [h_\sigma(x; y) * I(x; y)]$
 - b. $\nabla^2 h_\sigma(x; y) * I(x; y)$

- C) In a visual inference problem, we have some data set of observed features x , and we have a set of object classes $\{C_k\}$ about which we have some prior knowledge. Bayesian pattern classification asserts that: $P(C_k | x) = \frac{P(x | C_k)P(C_k)}{p(x)}$ [4 Marks]
- i. Give the name for, and explain the meaning of, each of these three terms:
 - a. $P(C_k | x)$
 - b. $P(x | C_k)$
 - c. $P(C_k)$
 - ii. Provide a decision rule using $P(C_k | x)$ for assigning classes based on observations in order to minimize misclassification.

Question 4

[12 Marks]

- A) Suppose an image has the following points : (1,1), (5,1), (2,2), (4,2), (3,3), (2,4), (4,4), (1,5), (3,5), (5,5).
- i. Show image space for these points and the corresponding lines in parameter space.
 - ii. Find the optimal values of parameter space.
 - iii. Draw the corresponding lines in image space.
- B) Implement Hough Transform using polar representation for image shown in part (A), consider $\theta (-45^\circ, 0^\circ, 45^\circ, 90^\circ)$.
- i. For each pixel (x, y) in image compute r as ($r = x \cos \theta + y \sin \theta$) and tabulate the results for different θ values.
 - ii. Show accumulator array which contains number of times each value of (r, θ) appears in table obtained in part (i).
 - iii. Determine largest values occur at (r, θ) and show the corresponding lines in image space.

Question 5

[8 Marks]

- A) Draw block diagram to show longest line detection (r_m, θ_m) using Hough Transform Algorithm (by using polar representation).
- B) Explain the Hough Transform algorithm used in part (A).
- C) Draw and explain all types of object tracking methods.

With Best Regards,
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